Subject Description Form

Subject Code	EEE523
Subject Title	Economics and Markets in Power Systems and Electrified Transportation
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To enable students to understand the key and practical issues of restructuring the electricity supply industry and to establish a broad knowledge of open electricity market operation. 2. To enable students to understand the key issues in open electricity market operation including deregulated power system operation, procurement of ancillary services, congestion management, so that students are provided with the knowledge and techniques they need to meet the electric industry's challenges in the 21st century. 3. To enable students to understand the current development and future progress of the electrical vehicle technology, including the advantages of EVs, the structure of EVs, the economy of EVs, major market players, and technical bottlenecks. 4. To enable students to understand the charging technology of EVs, the charging consideration of the EV users, the economic charging plan of EVs and corresponding incentive scheme, and optimization of EV charging infrastructure.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Acquire a good understanding of the rationale and key issues for restructuring the electricity supply industry, practical operation and design considerations for real world electricity markets, and financial tools to hedge risks used in electricity supply industries. b. Assess ancillary services requirements and values based on security, economic and performance considerations. c. Acquire a good understanding of the EV technology as well as their engagement in economic and green transportation. Assess the EV charging plan and infrastructure installation based on economic and performance considerations. d. Present technical results in the form of technical report and verbal presentation
Subject Synopsis/ Indicative Syllabus	1. ***Restructuring of the Electricity supply industry* (*ESI*):** ESI structures; Privatisation and competition; Market structures and architectures; Regulation of Electricity Markets; Role of existing players. 2. ***Electricity market*:** Timeline coordination, design considerations and practical operation of a real-world electricity market system. Use of different financial contracts/tools including derivatives and electricity futures for risk management in electricity markets. Game theory approach for market competition analysis. Transmission congestion management in the electricity market. Security considerations. 3. ***Ancillary services*:** Transmission ownership and restructuring. Measuring available transmission capacity in energy markets. Purchasing transmission capacity. Network and point to point transmission services. Fixed and firm transmission rights. Ancillary services and technical specifications, and performance based cost model. 4. ***Advancement of EV technology*:** The development history of EVs. Cost breakdown of EVs. Advantages of EVs. Major market players of EVs. Charging infrastructure of EVs.

5. ***Economic EV operation:*** Mileage anxiety issue of EVs. Charging incentive policies. Planning of charging station for EVs. Charging scheme of EVs.

Teaching/Learning Methodology

The concept of electricity and EV market modelling and economic analysis framework will be presented through lectures and tutorials with reference to real-life market environment. Students will be required to form groups to work through cases covering the market structure and operational aspects so as to develop ability to critically evaluate principles and operation of electricity and EV markets. Tutorials will be structured on different sessions for better understanding on the theoretical concepts which require sufficient contributions from students. Students will also learn through active participation in the presentation of finding of their case studies.

Teaching/Learning Methodology	Outcomes			
	a	b	c	d
Lectures	✓	✓	✓	
Case Studies & Presentation	✓	✓	✓	✓

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
		a	b	c	d
1. Examination	60%	✓	✓	✓	
2. In-class tests	18%	✓	✓	✓	
3. Assignments	10%	✓	✓	✓	
4. Mini-project and report	12%	✓	✓	✓	✓
Total	100%				

The outcomes on the concepts of modelling, analysis and applications are assessed by the usual means of examination and tests whilst those on problem-solving techniques and presentation of findings, as well as technical reporting and teamwork, are evaluated by the case study exercise.

Student Study Effort Expected

Class contact:	
▪ Lecture/Tutorial	33 Hrs.
▪ Presentation	6 Hrs.
Other student study effort:	
▪ Case study and report	15 Hrs.
▪ Self-study	51 Hrs.
Total student study effort	105 Hrs.

Reading List and References

Reference books:

1. D. Gan, D. Feng and J. Xie, Electricity Markets and Power System Economics, CRC Press, 2013
2. D. Kirschen, G. Strbac, Fundamentals of Power System Economics, 2nd Edition, John Wiley & Sons, 2018
3. K. Bhattacharya, M.H.J. Bollen, and J.E. Daalder, Operation of Restructured Power Systems, Kluwer Academic Publishers, 2001